CALVIN GACHERU MWANGI

Data Scientist & AI Engineer

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PROFESSIONAL SUMMARY

Results-driven Data Science and AI Engineering student with proven experience across the full machine learning lifecycle — from data engineering and model development to cloud deployment and MLOps automation. Specializes in building production-ready **Agentic AI** systems, **RAG (Retrieval-Augmented Generation)** architectures, and predictive ML models using **Python**. Skilled at translating raw data into scalable, business-ready software solutions with a strong focus on automation, CI/CD, and analytics-driven decision-making.

TECHNICAL SKILLS

Programming: Python (Advanced), SQL, R, Bash, HTML/CSS/JavaScript

Machine Learning & AI: scikit-learn, Pandas, NumPy, LangChain, LangGraph, RAG, Agentic AI, OpenAI API, Groq LLM, Qdrant, NLP, Time Series Analysis, Predictive Analytics

Infrastructure & MLOps: Docker, Kubernetes, CI/CD (GitHub Actions), PostgreSQL, Redis, FastAPI, Flask, Django, Linux, Git

Data & BI: ETL Pipelines, Exploratory Data Analysis, Data Visualization, Power BI, Matplotlib

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer — BikBat Insurance Agency *Nairobi, Kenya | 2026*

- Engineered and deployed a full-stack corporate web platform end-to-end for a commercial insurance client, delivering a production-ready digital presence aligned with business requirements.
- Directed the complete deployment lifecycle — including domain configuration, DNS management, and secure cloud hosting — ensuring a reliable, scalable launch with minimal downtime.

Data Science Virtual Intern — ProdigyInfoTech *Remote | 2025*

- Delivered 4 end-to-end data science projects spanning exploratory data analysis, predictive modeling, customer churn analytics, and NLP sentiment analysis within internship timelines.
- Built reproducible, well-documented data workflows that improved analytical transparency and reliability, emphasizing data quality and model interpretability across all deliverables.

SELECTED PROJECTS

MySaidi — AI Productivity Assistant (Agentic AI) *Python, FastAPI, LangGraph, Groq LLM, PostgreSQL*

- Architected a full-stack Progressive Web App (PWA) leveraging NLP and conversational AI to help users manage schedules and habits through natural language interaction.
- Designed and implemented a decoupled browser-to-REST API architecture, integrating a LangGraph-powered **Agentic AI** system driven by Groq LLMs to parse user intent and dynamically query/modify data via bound tool calling.

Uber Price Prediction & MLOps Pipeline *Docker, GitHub Actions, FastAPI, Streamlit, Random Forest*

- Engineered an enterprise-grade **MLOps** pipeline to predict ride prices using an optimized Random Forest Regressor, translating raw trip data into a deployable prediction service.
- Automated the full machine learning lifecycle — from training to deployment — using CI/CD pipelines (GitHub Actions) and multi-container Docker orchestration, reducing manual deployment overhead.

Diabetes Risk Screening Tool *XGBoost, SMOTE, scikit-learn, Streamlit*

- Developed a recall-optimized diagnostic pipeline to address severe class imbalance in medical screening data using SMOTE resampling and XGBoost.

- Achieved an **88% recall rate** for minority classes, prioritizing the minimization of false negatives in a medical diagnostics context.

Behavioural Explainer — RAG Architecture *PyMuPDF, Qdrant, Ollama, Llama 3.1*

- Constructed a production-grade **RAG** system to analyze and explain complex psychological framework scenarios using Llama 3.1.
- Engineered a document ingestion pipeline that mapped document embeddings into a Qdrant vector database, accessible through a custom Typer CLI for streamlined querying.

Lung Cancer Survival Prediction *Random Forest, Plotly, Survival Analysis*

- Collaborated on a data pipeline to extract and analyze clinical records, applying survival analysis techniques to estimate patient outcomes.
- Engineered advanced clinical features — including temporal tracking and comorbidity scoring — and deployed an interactive Plotly dashboard to visualize survival predictions.

EDUCATION

Bachelor of Science in Data Science & Analytics *Expected 2027*

United States International University Africa (USIU-Africa), Nairobi, Kenya — Year 4 of 4

CERTIFICATIONS

- Building AI Agents using LangGraph — Simplilearn (2026)
- Data Science Virtual Internship — ProdigyInfoTech (2025)